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General Comment

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Attachments

Middleton.Fed AI Action Plan

Expert Review of Public Comment on Federal AI Action Plan for Healthcare

Executive Summary:

This letter provides an expert-level public comment submitted in response to the Request for Information (RFI) for the development of a Federal Artificial Intelligence (AI) Action Plan.

The comment advocates for a multi-level AI governance framework, robust post-market surveillance mechanisms for AI, particularly in healthcare, the indispensable role of human oversight, the necessity of explainable AI recommendations, and the strategic integration of curated knowledge sources to optimize AI performance for licensed clinicians. The analysis presented makes recommendations that are strongly supported by current research and are critical for the safe, effective, and responsible integration of AI, especially within the healthcare domain.

Prioritizing these aspects in the Federal AI Action Plan will be essential for sustaining America's AI leadership while ensuring responsible innovation. Key recommendations include establishing a clear multi-level governance structure, mandating comprehensive post-market surveillance, emphasizing human oversight in AI applications, promoting the development and deployment of explainable AI, strategically integrating curated knowledge sources, and ensuring regulatory alignment across federal agencies.

Introduction:

The development of a Federal AI Action Plan is a significant national initiative aimed at fostering America's leadership in artificial intelligence while prioritizing responsible innovation. Public input, solicited through Requests for Information published in the Federal Register, plays a crucial role in shaping the strategic direction of this plan. This letter undertakes an expert review to make a public comment submitted in response to the RFI, focusing on its arguments concerning AI governance, post-market surveillance, human oversight, explainable AI, and the integration of curated knowledge, particularly within the healthcare sector. The purpose of this analysis is to provide a comprehensive evaluation of these critical considerations, drawing upon relevant research to inform policymakers involved in the formulation and implementation of the Federal AI Action Plan. By examining the evidence supporting the claims made in the public comment, this report seeks to contribute to the development of a robust and effective national strategy for AI in healthcare.

The Foundational Imperative for Multi-Level AI Governance:

This comment emphasizes the critical need for a well-defined and adaptable AI governance framework that operates cohesively across local, regional, and federal levels, especially within the diverse landscape of healthcare. While federal leadership is undoubtedly essential for setting overarching policies, ethical guidelines, and interoperability standards, the inherently varied contexts in which AI is deployed across different sectors, including healthcare, necessitate that governance also be deeply embedded at the local and regional levels. This multi-tiered approach ensures that the development and deployment of AI are not only guided by consistent overarching principles but are also responsive to the specific needs, values, and regulatory landscapes of different communities, healthcare systems, and research institutions. For instance, the application of AI in a rural healthcare setting may present different challenges and require different considerations compared to its use in a large urban academic medical

center. Governance mechanisms at the local level can be tailored to address these unique nuances more effectively than a purely centralized federal approach. Furthermore, the establishment of interoperability standards at the federal level is paramount. This would ensure that diverse AI systems utilized within different healthcare organizations can communicate and exchange data securely and efficiently, which is crucial for seamless patient care and effective health information management. Imagine a patient transitioning their care between healthcare providers in different states; their AI-driven health records should be readily accessible and understandable by the new providers, a scenario that necessitates common data standards and protocols established at the federal level.

Several sources corroborate the public comment's assertion regarding the current challenges in the AI landscape, including the absence of clear regulatory frameworks, the immaturity of existing governance structures, and inconsistencies in their application. These challenges underscore the urgency for the Federal AI Action Plan to provide much-needed clarity and structure to the currently fragmented AI governance environment. The lack of well-defined rules can create uncertainty for innovators and developers, potentially stifling progress, while inconsistencies in application can lead to uneven levels of safety and ethical considerations across different healthcare settings. A multi-level governance approach, as proposed here, is supported by research that suggests the necessity of developing a framework that aligns with ethics and fundamental human values, involving governments, corporations, and citizens as interdependent stakeholders. Examining the relationships between these groups through dimensions of trust, such as competence, integrity, and benevolence, provides practical insights for enhancing user experiences and informing public policy related to AI. This perspective adds the crucial dimension of trust between various stakeholders as a fundamental element for effective multi-level AI governance. If healthcare providers, patients, or the broader public lack trust in the AI governance mechanisms, they will be less inclined to adopt or rely on AI technologies, hindering their potential benefits.

AI governance, as defined by some research, encompasses a multi-layered approach that spans from organizational structure to regulatory alignment, emphasizing the establishment of standardization and regulatory frameworks to guide AI development and deployment. This reinforces the idea that multi-level governance needs to incorporate both top-down regulatory frameworks set by governmental bodies and bottom-up organizational practices implemented by individual healthcare providers and institutions. Federal regulations can establish the broad boundaries and requirements for AI use, but individual healthcare organizations need to develop their own specific policies and procedures to ensure compliance and foster responsible AI adoption within their unique operational contexts. An effective AI governance model can be further conceptualized through a three-level approach: operational implementation within business units, ethical decision-making and escalation through an ethics committee, and executive oversight at the senior management level. This provides a concrete structural model for how multi-level governance can function in practice within a healthcare organization. Operational teams directly using AI in their daily workflows can handle initial governance at their level, while more complex ethical concerns can be escalated to a dedicated committee for thorough evaluation, and ultimately, the executive board provides overall strategic direction and accountability for AI governance across the organization. While the primary focus of the Federal AI Action Plan is domestic, it is important to acknowledge that AI governance has international dimensions and requires cooperation across different sectors. These sources emphasize the need for multi-stakeholder and multi-level collaboration in managing the global impacts of AI, proposing a framework that involves governments, corporations, and citizens. Therefore, the US AI Action Plan should consider international standards and foster collaboration with other nations on AI governance to ensure a cohesive global approach to this transformative

technology.

To effectively address the challenges posed by the evolving AI landscape, the AI Action Plan should prioritize a clear delineation of responsibilities and jurisdictional authority across federal agencies, state and local governments, and individual organizations involved in AI development and deployment. This includes explicitly defining the roles of regulatory bodies such as the FDA in the context of healthcare AI [1, 6-7], as well as the responsibilities of healthcare providers and institutions in ensuring the safe and ethical usage of AI. Clearly defining the FDA's role is particularly critical given the increasing proliferation of AI-enabled medical devices and the need for robust regulatory oversight to ensure their safety and effectiveness. Ambiguity regarding which agency is responsible for overseeing specific aspects of AI in healthcare can lead to regulatory gaps, delays in approvals, and potential risks to patient safety. The evolving AI landscape presents several inherent challenges, including algorithmic bias, the often-opaque nature of AI decision-making (the "black box"), and the risks associated with uncontrolled or poorly governed AI systems, often termed "shadow AI". These challenges underscore the necessity for clearly defined responsibilities for monitoring, auditing, and mitigating these risks at all levels of the governance framework. Without a designated entity or individual accountable for checking for and addressing bias in AI algorithms, for example, these biases can easily go undetected and potentially lead to discriminatory outcomes or harm to patients.

Furthermore, the development and implementation of AI governance face global challenges such as fragmented disciplinary collaboration, ineffective multilateral coordination, and disconnects between policy design and grassroots implementation. The US Federal AI Action Plan needs to proactively address these potential pitfalls to ensure its effectiveness in the US context. Policies developed in isolation, without meaningful input from experts across various disciplines or without careful consideration of the practicalities of implementation at the local level, may ultimately fail to achieve their intended goals. The need for AI governance to address critical aspects such as compliance with regulations, ethical considerations, and fostering public trust is also paramount. Principles and frameworks developed by organizations like NIST, OECD, and the European Union provide valuable guidance in this regard, and the Action Plan should aim to align with these established standards to promote consistency and avoid redundant efforts. In the specific context of US healthcare, several AI governance frameworks and guidelines have been proposed or implemented, highlighting the need for adaptive strategies, robust oversight around potential biases, and strong engagement from both healthcare leadership and regulatory bodies. These healthcare-specific initiatives offer concrete examples and principles that can inform the structuring of AI governance within the broader federal plan, recognizing the unique risks and ethical considerations inherent in the healthcare domain.

Recognizing the dynamic nature of AI technology, policymaking and governance structures must incorporate mechanisms for continuous evaluation, iterative review, and timely revision. Rigid, pre-defined controls may quickly become outdated or ill-suited to address new capabilities and potential risks. The regulatory framework should therefore embed feedback loops that capture the real-world consequences of AI deployment, allowing for adaptive adjustments to policies and guidelines. This necessitates the development of novel evaluation methodologies that can keep pace with the rapid evolution of AI while maintaining rigorous standards for safety, efficacy, and fairness. Expert consensus, as demonstrated in the development of digital health competency frameworks and AI guideline recommendations, provides a valuable model for establishing such adaptable frameworks [1, 13-14]. The inherently dynamic nature of AI technology necessitates that its governance framework be a "living" document, capable of evolving and adapting in response to new technological advancements,

emerging risks, and real-world experiences. Unlike traditional regulations designed for more static technologies that are designed for more static technologies, AI governance requires built-in mechanisms for regular updates and revisions to remain relevant and effective. The inclusion of continuous monitoring mechanisms and feedback loops is crucial for refining governance structures over time. Regular audits and assessments are essential to ensure the ongoing effectiveness and relevance of AI governance practices, allowing for adjustments based on the actual impact and performance of AI systems in healthcare settings. Implementing proactive AI governance measures early on can also provide a significant competitive advantage for organizations and ensure their long-term sustainability [2]. By establishing clear guidelines and responsibilities from the outset, healthcare organizations can mitigate potential risks, enhance public trust, improve efficiency, and foster a culture of responsible innovation. Waiting for regulations to become fully established and potentially rigid may make it more challenging for organizations to adapt their practices later.

Establishing Robust Strategies for Post-Market Surveillance of AI and Agents in Healthcare:

Given the capacity of AI systems, particularly those powered by Large Language Models (LLMs), to learn, adapt, and potentially evolve their behavior after deployment, a regulatory approach solely focused on pre-market evaluation is insufficient to ensure ongoing safety and effectiveness. Drawing parallels with the well-established post-market surveillance practices for pharmaceuticals and traditional medical devices, the AI Action Plan must prioritize the development and rigorous implementation of comprehensive post-market surveillance mechanisms for AI and AI agents in all sectors, especially healthcare. This analogy with the pharmaceutical and medical device industries is critical for understanding the necessity of ongoing monitoring of AI in healthcare. Just as the long-term effects and safety of a new drug are carefully tracked after it is released to the market, the performance, safety, and potential for evolving behavior of AI systems in healthcare must also be continuously monitored in real-world use. Established post-market surveillance (PMS) practices in these domains, overseen by regulatory bodies like the FDA, involve systematic data collection, analysis, and reporting to detect adverse events and ensure the continued safety and effectiveness of products. These established practices provide a wealth of knowledge and experience that can inform the development of similar mechanisms tailored for AI in healthcare. For medical devices specifically, regulations like 21 CFR Part 822 outline the requirements for PMS, emphasizing the importance of real-world data, user feedback, and manufacturer responsibility for ongoing surveillance [23-24]. The principles of risk-based surveillance, where higher-risk devices undergo more stringent monitoring, and the active collection of data from various sources, including users and healthcare professionals, are useful for establishing PMS for AI-enabled medical devices [25-28].

Key strategies for effective post-market surveillance should include:

- Standardization of Governance and Adverse Event Reporting:** Establishing standardized protocols for defining, identifying, collecting, and systematically monitoring AI-related safety events and potential harms, in close collaboration with patient safety organizations, healthcare providers, and regulatory agencies, is crucial. This will enable the systematic tracking of both the intended benefits and any unintended negative consequences of AI deployment in real-world settings. The development of a common reporting framework for AI incidents, as suggested by the OECD, provides a valuable starting point. Standardization is paramount for enabling effective aggregation and analysis of data related to AI performance and safety across diverse healthcare environments. Without common definitions for AI-related adverse events and standardized reporting formats, it will be exceedingly difficult to identify overarching trends and patterns that could indicate systemic issues or emerging risks associated

with AI deployment. The EU AI Act also mandates post-market monitoring for high-risk AI systems, requiring providers to actively collect and analyze data on their performance throughout their lifecycle. This international regulatory trend further underscores the importance of establishing robust PMS mechanisms for AI in the US.

- **Continuous Performance Monitoring and Evaluation:** Implementing robust mechanisms for the ongoing monitoring and evaluation of AI system performance after deployment is essential. This includes tracking key performance indicators (KPIs), identifying potential biases that may emerge or become amplified over time ("bias in, bias out"), and continuously assessing the clinical utility, safety, and effectiveness of the AI in diverse patient populations and real-world clinical workflows. Continuous monitoring is vital because the performance of AI systems can fluctuate over time due to various factors, such as shifts in the underlying data (data drift) or updates to the AI model itself. A pre-market assessment of an AI system's performance provides only a snapshot in time and may not accurately reflect its long-term behavior or effectiveness in the complex and ever-changing real-world healthcare environment. Research argues strongly for more robust post-market surveillance of AI in healthcare, noting the limitations of current FDA frameworks that primarily focus on premarket evaluation [31]. The FDA's current approach may not adequately address the practical challenges of continuously monitoring AI performance across diverse patient populations and clinical settings.
- **Mandatory AI Use Reporting and Comprehensive Documentation:** Establishing clear standards for AI use reporting and comprehensive documentation, including detailed metadata on the underlying algorithms, training data provenance, model architecture, intended use cases, and performance metrics, is vital for ensuring transparency, accountability, and the ability to trace the AI's behavior. This documentation should ideally break down impacts by specific model use phases and clearly defined use cases. Comprehensive documentation serves as a foundational element for transparency and accountability in the use of AI in healthcare. Detailed records of an AI system's design, training data, performance metrics, and intended applications are essential for understanding how the system functions and for investigating any potential issues or adverse events that may arise. This information is also crucial for regulatory oversight and for facilitating independent audits of AI systems. While AI itself can be leveraged to aid in post-market surveillance, including automating the generation of regulatory documents and analyzing real-world data, the establishment of clear and standardized reporting requirements remains paramount. AI tools can help manage and analyze the vast amounts of data generated through PMS, but well-defined guidelines are necessary to ensure that the right types of data are collected and reported in a consistent manner.
- **Effective Incident Reporting and Remediation Systems:** Establishing clear and accessible pathways for clinicians, patients, and other stakeholders to report incidents, adverse events, or near misses associated with AI use is paramount. Learning from these reported incidents will be crucial for iterative improvements to AI systems, the development of effective risk mitigation strategies, and the implementation of necessary corrective actions. Easy and reliable mechanisms for reporting AI-related incidents are essential for capturing a complete understanding of the technology's impact in real-world clinical practice. If the reporting process is cumbersome, unclear, or if there are concerns about potential repercussions for reporting, many incidents, particularly near misses or subtle issues, may go unreported. Establishing a culture of open reporting, where healthcare professionals and patients feel comfortable raising concerns without fear of penalty, is crucial for effective post-market surveillance. The information gleaned from these reports can then be used to drive iterative improvements to AI systems, develop more effective strategies for mitigating identified risks, and implement necessary corrective actions to prevent future occurrences.

- Periodic Independent Auditing and Validation:** Implementing a system for periodic auditing and independent validation of deployed AI systems, potentially conducted by qualified third-party experts, can provide an objective assessment of their ongoing safety, effectiveness, fairness, and adherence to evolving regulatory standards. Independent audits offer an unbiased perspective on the performance and safety of AI systems, potentially identifying issues or areas of concern that internal teams might overlook due to familiarity or inherent biases. Engaging qualified third-party experts to conduct these audits can further enhance their objectivity and credibility. These periodic assessments can evaluate not only the technical performance of the AI system but also its adherence to evolving regulatory requirements, ethical guidelines, and standards for fairness and equity. This added layer of scrutiny can provide greater assurance to patients, healthcare professionals, and regulators regarding the trustworthiness and responsible use of AI in healthcare.
- Active Solicitation and Integration of User Feedback:** Actively soliciting and thoughtfully incorporating feedback from clinicians, other healthcare professionals, and patients on their experiences with AI tools is essential for identifying areas for improvement, uncovering potential risks or usability issues, and ensuring that AI systems effectively meet the needs of end-users in diverse clinical contexts. Feedback from the individuals who directly interact with AI systems in their daily clinical practice or as patients receiving care provides invaluable insights into the real-world usability, effectiveness, and potential challenges associated with these tools. Actively seeking out and carefully considering this feedback is crucial for identifying areas where AI systems can be improved, uncovering previously unforeseen risks or usability issues, and ultimately ensuring that these technologies are effectively meeting the needs of end-users across diverse clinical settings and patient populations. This user-centered approach is essential for the successful and responsible integration of AI into healthcare workflows.

The Essential Roles of Human Oversight in AI-Driven Healthcare:

This comment highlights the critical importance of human oversight in the development and implementation of AI in the healthcare sector. Research strongly supports the benefits of incorporating human intelligence and judgment into AI systems, often referred to as Human-in-the-Loop (HITL) approaches. HITL AI is characterized by continuous human involvement throughout the AI lifecycle, including data annotation, model training, validation, and real-time operation. This ongoing interaction allows for iterative learning, where the AI system evolves by incorporating feedback from human experts, refining its algorithms and enhancing its ability to handle complex or ambiguous situations. Furthermore, human oversight is crucial for ensuring that AI decisions adhere to ethical standards and societal norms, mitigating the risk of bias and unintended consequences that may arise from fully autonomous systems. The integration of human expertise with the speed and precision of machine learning creates a collaborative synergy that aims to produce more accurate, adaptable, and ethical AI solutions. In the context of healthcare, skilled healthcare professionals acting as the "human in the loop" are indispensable for guiding AI systems effectively and ethically. While AI can analyze vast amounts of data and identify patterns, it often lacks the clinical intuition, empathy, and nuanced understanding of individual patient circumstances that only a human can provide. This collaboration between human expertise and AI capabilities leads to better and more precise outcomes, ensuring that AI enhances rather than replaces the critical human element in healthcare.

The benefits of human oversight in AI extend to several key areas. It ensures ethical decision-making, where humans define ethical guidelines and review AI outputs for biases and discrimination. Human oversight also fosters accountability, ensuring transparency in AI systems and providing a mechanism for addressing errors [36]. Moreover, humans provide adaptability and contextual understanding, complementing AI's analytical power with the ability to navigate dynamic situations and consider diverse perspectives [36]. This continuous feedback loop, where humans intervene to

correct errors and refine outputs, enhances AI learning and improves the accuracy and reliability of algorithms over time [38]. In healthcare specifically, HITL AI can aid in medical diagnostics and treatment planning, with human doctors validating and refining AI-generated results to ensure accuracy and safety. This human validation is critical in a field where mistakes can have severe consequences. The complexities of human involvement in AI systems require careful consideration, including differentiating between having a human directly "in the loop" versus "near the loop," and the importance of involving patients in the governance of AI systems that affect their care [39]. While persistent human involvement in every AI decision may not be scalable, ensuring human oversight at critical junctures and incorporating patient perspectives are essential for responsible AI implementation. Research shows that human oversight in AI medical diagnosis and treatment is crucial for ethical application, clinical intuition, unique perspectives, skill refinement, responsibility, control, and reducing patient harm.

Conversely, there are significant risks associated with over-reliance on fully autonomous AI in healthcare without adequate human oversight. These risks include the potential for errors in diagnosis or treatment recommendations, breaches of patient privacy due to compromised AI systems, the perpetuation or amplification of biases present in the training data leading to unequal care, a lack of the essential personal touch and empathetic understanding that human practitioners offer, and the danger of diagnostic over reliance where clinicians may become overly dependent on AI and overlook critical aspects of a patient's condition. Furthermore, errors made by AI systems can propagate if the AI continues to learn from incorrect data, reinforcing these mistakes. Unequal access to sophisticated AI technology could also widen existing health disparities. The ethical considerations surrounding the use of fully autonomous AI in life-and-death decisions and the complexities of assigning accountability in case of AI-driven errors are also significant concerns [42-45]. Therefore, it is crucial to maintain human clinical judgment as central to patient care [8]. AI should serve as a tool to support and augment the expertise of healthcare professionals, not as a replacement for their critical thinking, nuanced understanding, and ethical considerations [46]. The ultimate responsibility for patient well-being must remain with human clinicians, who can interpret AI-generated information within the broader context of a patient's individual needs and circumstances.

The Necessity for Explainable AI Recommendations in Clinical Practice:

This comment emphasizes the necessity for AI recommendations in clinical practice to be explainable. Explainable AI (XAI) is a field dedicated to making AI models more interpretable and transparent, ensuring that their decision-making processes can be understood and trusted by clinicians and other stakeholders. This is particularly crucial in healthcare, where the rationale behind a diagnosis or treatment recommendation can have profound implications for patient outcomes [48-50]. XAI aims to overcome the "black box" problem associated with many advanced AI models, such as deep learning networks, whose intricate workings are often opaque even to their developers. By providing insights into how an AI system arrives at a particular conclusion, XAI fosters trust among clinicians, who are more likely to adopt and rely on AI tools if they can understand the reasoning behind their recommendations [51]. Furthermore, explainability enhances transparency in healthcare processes, allowing for greater scrutiny and validation of AI-driven insights [52-55].

The implementation of XAI in healthcare presents both challenges and significant benefits. Challenges include the need for extensive and high-quality datasets to train AI models effectively, the substantial computational resources required for many XAI techniques, and the inherent difficulties in deploying XAI solutions in real-world clinical settings due to data quality and availability issues [51]. Additionally, there is often a trade-off between the accuracy of an AI model and its interpretability; highly accurate complex models may be less explainable, while more interpretable models might sacrifice some predictive performance [56-58]. Ensuring legal and regulatory compliance, particularly with data privacy regulations, also poses a challenge for XAI implementation

[59-61]. Despite these hurdles, the benefits of XAI in healthcare are substantial [62]. XAI can improve diagnostic accuracy by allowing clinicians to understand the factors that contributed to an AI's diagnosis, potentially revealing previously unrecognized relationships between symptoms and diseases [63-65]. It also plays a crucial role in advancing personalized treatment plans by providing interpretable insights into individualized recommendations, clarifying the criteria behind suggested therapies or medications based on a patient's unique genetic, environmental, and lifestyle factors [66-68]. By making AI decision-making processes transparent, XAI enhances accountability, allowing clinicians to identify potential errors or biases in AI outputs and ensuring that AI systems complement rather than replace human judgment [49]. The necessity for explainable AI in clinical practice stems from the fundamental need for clinicians to understand and trust the tools they use to make critical decisions about patient care.

The Strategic Importance of Integrating Curated Knowledge Sources to Optimize AI Performance for Licensed Clinicians:

This comment also highlights the strategic importance of integrating curated knowledge sources to optimize the performance of AI systems used by licensed clinicians. Research indicates that AI benefits significantly from access to high-quality, verified information, leading to improved accuracy and a reduction in the generation of incorrect or nonsensical outputs, often referred to as hallucinations [69-72]. Curated knowledge bases, which are carefully selected and organized collections of authoritative information, can provide AI systems with a reliable and up-to-date source of medical knowledge, thereby enhancing the trustworthiness and utility of their recommendations for healthcare professionals [73-77]. The integration of such knowledge sources can lead to more accurate diagnoses, more effective treatment planning, and better overall patient outcomes.

The benefits of using curated knowledge sources in AI for healthcare extend beyond just improving accuracy [78-80]. These sources can also facilitate automated updates to AI systems with the latest medical findings, improve the efficiency of knowledge retrieval and analysis, and enhance the integration of AI with existing healthcare support systems [81-84]. By providing AI with access to a centralized and validated repository of medical information, healthcare organizations can ensure that clinicians are receiving recommendations based on the most current evidence-based practices [85-87]. Techniques like Retrieval-Augmented Generation (RAG) play a crucial role in leveraging curated knowledge [78]. RAG is a method that enhances the capabilities of Large Language Models by allowing them to access and incorporate information from external, curated data sources in real-time [79]. In healthcare, RAG can enable clinicians to ask complex medical questions and receive accurate, contextually relevant answers that are grounded in the latest research, clinical guidelines, and patient data [80]. This approach can significantly improve clinical decision support, accelerate medical research, enhance patient engagement through better communication, and streamline administrative tasks within healthcare organizations [88-93].

Furthermore, the integration of curated knowledge sources, such as knowledge graphs and verified databases, is strategically important for mitigating the risks of data poisoning, where malicious actors introduce false information into AI training data [71]. By grounding AI outputs in trusted and curated knowledge, healthcare organizations can increase the reliability and trustworthiness of AI recommendations, safeguarding against the propagation of misinformation that could potentially harm patients. Knowledge graph-based agents, for example, can integrate non-codified knowledge from LLMs with structured knowledge from medical concept graphs to improve the accuracy and specificity of medical reasoning and question answering. This approach acts as a "truth filter," helping to identify and prevent the spread of inaccurate or harmful information, thereby increasing confidence in the use of AI in mission-critical healthcare settings.

Navigating the Regulatory Landscape for AI in US Healthcare:

The FDA's recent efforts, including the release of draft guidance documents and action plans, indicate a proactive approach to addressing the complexities of AI in healthcare [94-112]. These initiatives highlight the agency's recognition of the need for a flexible and adaptive regulatory framework that can keep pace with the rapid advancements in AI technology. The FDA's focus on the Total Product Lifecycle (TPLC) approach is particularly relevant for AI, as it emphasizes the importance of ongoing monitoring and evaluation of AI systems after they are deployed. This aligns with the earlier discussion on the necessity of robust post-market surveillance mechanisms.

Furthermore, the regulatory landscape for AI in healthcare extends beyond the FDA. Other agencies, such as the Office of the National Coordinator for Health Information Technology (ONC), also play a crucial role in shaping the regulatory environment. The ONC's Health Data, Technology, and Interoperability (HTI-1) Final Rule, for example, includes provisions related to the transparency of AI-powered decision support interventions in Electronic Health Records (EHRs) [11]. These regulations underscore the need for coordination and alignment across different agencies to ensure a cohesive and effective approach to AI governance in healthcare.

The emergence of Large Language Models (LLMs) and other forms of generative AI has added another layer of complexity to the regulatory landscape [113-116]. These technologies possess unique capabilities and potential risks that may require different regulatory strategies compared to traditional AI-based medical technologies. The potential for LLMs to generate misinformation or biased outputs, for instance, necessitates careful consideration of how these systems are validated and monitored. The imperative for regulatory oversight is particularly pronounced for LLMs in healthcare, given their potential to impact clinical decision-making and patient safety.

Conclusion and Recommendations: Prioritized Action Items:

This analysis of the relevant research and related policies reveals a strong consensus on the critical importance of the issues raised. The need for multi-level AI governance, robust post-market surveillance, essential human oversight, explainable AI recommendations, and the strategic integration of curated knowledge sources are all vital for ensuring the safe, effective, and responsible adoption of AI in healthcare. These issues are interconnected, but some require more immediate attention and foundational work to enable others.

To effectively address these imperatives and to guide the development of the Federal AI Action Plan, the following recommendations are proposed, *presented in order of priority, with designated responsible entities*:

I. Foundational Pillars (Highest Priority):

1. Establish a Clear Federal AI Governance Framework for Healthcare (Priority 1):

- **Action:** Define the roles, responsibilities, and jurisdictional authority of federal agencies (specifically including the FDA, HHS, ONC, NIST, and potentially a new dedicated AI oversight body) in regulating AI in healthcare. This framework must address pre-market evaluation, post-market surveillance, data privacy, algorithmic bias, and liability. It should explicitly address the unique challenges of Large Language Models (LLMs) and other generative AI technologies in healthcare.
- **Target Entity:** Congress (to enact legislation), HHS (to lead interagency coordination), FDA (to develop specific guidance for AI/ML as a medical device), ONC (to address AI in health IT and interoperability), NIST (to develop standards and best practices for AI safety and trustworthiness).
- **Rationale:** This is the highest priority because a clear, comprehensive federal framework is essential for all other actions. Without it, there's a risk of regulatory

gaps, inconsistencies, and stifled innovation. This framework must be established *before* widespread deployment of AI in high-stakes healthcare applications.

2. Develop and Mandate Standards for Post-Market Surveillance of Healthcare AI (Priority 2):

- **Action:** Create mandatory reporting requirements for AI-related adverse events in healthcare, mirroring existing systems for pharmaceuticals and medical devices. This includes standardized definitions of "adverse event" and "near miss" specific to AI, along with clear reporting pathways for clinicians, patients, and healthcare organizations. Develop protocols for continuous performance monitoring, including metrics for accuracy, bias, and clinical utility.
- **Target Entity:** FDA (primary responsibility for medical device and pharmaceutical surveillance), HHS (to coordinate with patient safety organizations), and a potential public-private collaborative (including healthcare providers, AI developers, and patient advocacy groups) to develop reporting standards.
- **Rationale:** Post-market surveillance is crucial for identifying and mitigating risks that may only emerge after deployment. This needs to be established early to capture data on the real-world performance and safety of AI systems.

3. Promote and Incentivize the Development and Adoption of Explainable AI (XAI) in Healthcare (Priority 3):

- **Action:** Fund research into XAI techniques specifically tailored for healthcare applications. Develop guidelines and best practices for implementing XAI in clinical settings. Consider incorporating XAI requirements into regulatory approval processes for high-risk AI systems. Create educational resources for clinicians on interpreting and using XAI outputs.
- **Target Entity:** NIH (to fund research), NIST (to develop standards and guidelines), FDA (to consider XAI in regulatory approvals), professional medical organizations (to develop educational resources), and academic institutions.
- **Rationale:** Explainability is essential for building trust and ensuring that clinicians can understand and validate AI recommendations. This will facilitate adoption and responsible use of AI in clinical practice.

II. Enabling Actions (Medium Priority):

4. Foster the Integration of Curated Knowledge Sources into Healthcare AI Systems (Priority 4):

- **Action:** Support the development and maintenance of high-quality, curated and computable (interoperable) medical knowledge bases (e.g., knowledge graphs, verified clinical guidelines). Incentivize AI developers to utilize these resources (e.g., through grants, partnerships, or regulatory requirements). Develop standards for interoperability between AI systems and curated knowledge sources. Promote and provide training resources for Retrieval-Augmented Generation (RAG).
- **Target Entity:** NIH (to fund research and development of knowledge bases), NIST (to develop interoperability standards), professional medical organizations (to curate and maintain knowledge resources), and private sector AI developers.

- **Rationale:** Curated knowledge sources are essential for improving the accuracy, reliability, and safety of AI systems, particularly in mitigating the risk of "hallucinations" and data poisoning.
5. **Establish Mechanisms for Meaningful Human Oversight of AI in Healthcare (Priority 5):**
- **Action:** Develop guidelines and best practices for implementing Human-in-the-Loop (HITL) approaches in various healthcare settings. These guidelines should address ethical considerations, workflow integration, and the allocation of responsibility between humans and AI. Provide training for healthcare professionals on how to effectively interact with and oversee AI systems.
 - **Target Entity:** FDA (to incorporate HITL considerations into regulatory guidance), professional medical organizations (to develop guidelines and training programs), healthcare institutions (to implement HITL workflows), and AI developers (to design systems that support human oversight).
 - **Rationale:** While human oversight is crucial, its implementation needs to be carefully considered to avoid creating bottlenecks or undue burden on clinicians. Clear guidelines and training are essential for effective and ethical integration of human oversight.

III. Supporting and Sustaining Actions (Ongoing Priority

1. **Data Governance and Quality Initiatives (Ongoing):**
 - **Action:** Invest in initiatives to improve the quality, representativeness, and accessibility of healthcare data used to train and validate AI systems. Establish standards for data curation, labeling, and sharing. Promote the use of federated learning and other privacy-preserving techniques to enable data sharing while protecting patient confidentiality.
 - **Target Entity:** ONC (to develop data standards and promote interoperability), NIH (to fund research on data quality and bias mitigation), healthcare organizations (to implement data governance policies), and data science communities.
 - **Rationale:** High-quality data is essential for the development of reliable and unbiased AI systems. Addressing data quality issues and promoting responsible data sharing practices will be crucial for the long-term success of AI in healthcare.
2. **Ethical Considerations and Bias Mitigation (Ongoing):**
 - **Action:** Establish an ongoing advisory body or task force dedicated to addressing ethical considerations related to AI in healthcare, including algorithmic bias, patient privacy, and informed consent. Develop tools and methodologies for detecting and mitigating bias in AI algorithms. Promote research on the ethical implications of AI in healthcare.
 - **Target Entity:** HHS (to establish the advisory body), NIH (to fund research), FDA (to incorporate ethical considerations into regulatory guidance), and academic institutions (to conduct ethical research and develop bias mitigation tools).
 - **Rationale:** Ethical considerations are paramount in healthcare AI. Ongoing attention and expertise are needed to address emerging ethical challenges and ensure that AI systems are used in a fair, equitable, and responsible manner.
3. **Workforce Development and Training (Ongoing):**
 - **Action:** Invest in training programs to equip healthcare professionals with the skills and knowledge needed to effectively use and oversee AI systems. Develop curricula for medical, nursing, and other healthcare education programs that address AI literacy and competency. Support initiatives to train data scientists and AI developers in healthcare-specific considerations and ethical principles.

- **Target Entity:** HRSA (to fund training programs), professional medical organizations (to develop curricula), academic institutions (to integrate AI into education), and AI developer communities.
 - **Rationale:** The successful integration of AI into healthcare requires a workforce that is comfortable and competent in using these technologies. Ongoing training and education are essential to ensure that healthcare professionals can effectively collaborate with AI systems and make informed decisions.
4. **Patient Engagement and Empowerment (Ongoing):**
- **Action:** Develop resources and initiatives to educate patients about AI in healthcare and empower them to participate in decisions about its use in their care. Promote transparency and open communication about how AI is being used in clinical settings. Support patient advocacy groups in representing the interests of patients in AI governance discussions.
 - **Target Entity:** AHRQ (to develop patient education materials), patient advocacy groups, healthcare organizations (to promote transparency), and community health centers.
 - **Rationale:** Patients are key stakeholders in the use of AI in healthcare. Engaging and empowering patients will build trust, ensure that their values are considered, and promote the responsible use of AI.

By prioritizing and implementing these recommendations in a structured and ongoing manner, the Federal AI Action Plan can effectively navigate the complex landscape of AI in healthcare. This comprehensive approach will foster innovation, ensure patient safety, and uphold ethical principles, ultimately leading to better health outcomes for all Americans. The continued dedication to these actions and the collaborative efforts of all stakeholders will solidify the United States' leadership in the responsible development and deployment of AI in healthcare.

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